

MEMS SENSORS VS PRINTED NANO SENSORS

Aspect	MEMS (Micro-Electro-Mechanical Systems) Sensors	Printed Nano Sensors
Fabrication method	Silicon-based microfabrication techniques	Additive manufacturing/printing techniques
Primary material	Silicon acts as both a structural and a sensing material	Nanomaterials, metals, or composite inks
Sensing mechanism	Often piezoresistive: doped silicon changes resistance under stress	Depends on engineered inks/materials designed for strain or force sensitivity
Structural role of material	The same silicon layer provides both structure and sensing	The transducer material can be freely chosen
Design approach	Performance tuned via microscale geometry (cantilevers, membranes, etc)	Performance tuned via material composition, low Young's modulus materials and placement on surfaces
Sensitivity control	Strongly dependent on silicon properties and device geometry	More dependent on functional ink properties and strain localisation
Key limitation	Constrained by intrinsic silicon properties and fabrication limits	Material stability, consistency, and long-term reliability can be challenges
Key advantage	High precision, mature, and well-controlled manufacturing	High flexibility in materials and integration with diverse surfaces
Strain detection strategy	Distributed within engineered microstructures	Often localised at high-strain surface regions
Typical applications	Pressure sensors, inertial sensors, automotive and industrial sensing	Flexible electronics, bio-integrated devices, ultra-thin sensors

For example, in the automotive industry, some companies intend to use these embedded sensors for wheel and axle alignment and to measure vehicle weight distribution. Conventional systems rely on bulky equipment, whereas the embedded approach offers a smaller, more easily integrated alternative. In other applications, the sensors are used to check and calibrate force in diagnostic equipment, where highly accurate measurements are essential.

Different industries are adopting the technology for different reasons. In medical applications, development cycles are longer, but the long-term value is significant because of the need for extremely accurate and reliable measurements. In industrial sectors such as manufacturing and automotive, adoption is faster because companies can more quickly realise practical benefits and typically have shorter product development cycles.

Some large manufacturers are already integrating this technology into automotive components to make them sensitive to force, allowing the components to directly report the loads and stresses they experience.

Small sensors, big problems!

While nanotechnology has long promised smaller, more cost-effective, and energy-efficient sensors, most

developments remain at the laboratory proof-of-concept stage rather than achieving full industrial-scale deployment. The transition from prototype to mass production is constrained by a combination of technical, economic, and organisational barriers. The main barriers to scaling can be summarised in four key areas:

- **Fabrication and scaling complexity.** As sensor dimensions shrink, fabrication becomes significantly more complex, requiring advanced techniques like lithography or nano-imprinting. Although these methods are established, they can increase process complexity, capital requirements, and non-recurring engineering effort, making large-scale production challenging.

- **Manufacturing, integration, and supply chain fragmentation.** Nanosensor production involves multiple tightly linked stages such as substrate preparation, metallisation, structuring, sensor fabrication, and packaging (contacting, sealing, cabling, and readout integration). Because these steps are often distributed across different suppliers, coordination challenges and supply chain vulnerabilities arise, often requiring partial vertical integration.

- **Market adoption and organisational resistance.** Even when performance advantages are clear, many organisations are reluctant to adopt highly innovative technologies

due to perceived risk and integration complexity. Multiple internal approval layers and difficulty in communicating business value further slow down decision-making and large-scale adoption.

- **Collaboration, IP complexity, and commercialisation barriers.**

Traditional intellectual property arrangements can slow down partnerships and development cycles. To reduce friction, simplified pilot-based models are often used, enabling customers to test the technology at a small scale after covering non-recurring engineering costs, supported by strong patent portfolios to streamline agreements and accelerate market entry.

To address these challenges, companies are increasingly bringing more manufacturing steps in-house to reduce dependency on external suppliers, adopting simplified pilot-based engagement models that allow customers to validate performance through small-scale deployments, leveraging strong patent portfolios to streamline intellectual property discussions, and planning for scalable production capacities (e.g., around 10,000 sensors per day) to support a smooth transition to industrial-scale rollout.

Scaling nanosensor production

Scaling nanosensor production in practice requires a complete shift